



AI-Driven Soil Sampling & Carbon Modelling Module

Detailed Functional Specification — Lean Edition

VM0042 v2.2	QA1 — Measure & Model	QA2 — Measure & Remeasure	DNDC · DayCent	AI Agent
+ MCP				

Prepared for: Carbonature Engineering & AI development team
Owner: Nitzan Bauer · Carbonature
Document version: 1.0 · April 2026
Audience: Lead developer / AI tooling engineer

1. Document control

Field	Value
Document title	AI-Driven Soil Sampling & Carbon Modelling Module — Detailed Specification (Lean)
Project	Carbonature platform — VM0042 module
Initial requirements doc	"איפיון ראשוני - מודול מונחה בינה מלאכותית לסבבי דיגום קרקע" (Carbonature, internal)
Reference standards	VCS VM0042 v2.2 · VMD0053 · VT0014 · ISO 11272:2017 · ISO 18400-104:2018 · ISO/IEC 17025
Carbon models in scope	DNDC v9.x (primary) · DayCent v6.x

Cloud target	AWS (eu-west-1 default)
Map engine	Mapbox GL JS
Form factor	Web only (no native mobile in v1)
Document length target	Lean (~25–35 pages)

Glossary

Term	Meaning
VM0042	VCS methodology for Improved Agricultural Land Management (v2.2 active 21-Oct-2025)
QA1 / QA2	Quantification Approach 1 (Measure & Model) / Approach 2 (Measure & Remeasure)
SOC	Soil Organic Carbon
BD	Bulk Density (g/cm ³)
ESM	Equivalent Soil Mass (mandatory reporting basis)
WP	"With Project" — project polygon / sampling unit
BSL	Baseline Control Site (QA2)
MVR	Model Validation Report (QA1, per VMD0053)
IME	Independent Modeling Expert (reviews MVR)

MCP	Model Context Protocol — Anthropic open standard. Used to connect the AI Agent to field tools and external systems.
MDD	Minimum Detectable Difference (FAO 2019 power-analysis output)
VCU	Verified Carbon Unit (1 t CO ₂ e)

2. Executive summary

The module is a **web-based, AI-augmented operating layer** for managing soil sampling cycles, ingesting lab data, running carbon models (DNDC, DayCent), and producing audit-ready outputs aligned with VM0042 v2.2 — across all Carbonature agricultural carbon projects.

It serves four distinct user types: a **Super Admin** with full control, an **AI Agent** that proposes plans, executes model runs, and answers questions, an **MRV Manager** who plans cycles and approves agent recommendations, and a **Soil Sampler** (external contractor) connected via the Model Context Protocol (MCP) for in-field data capture.

The platform supports **both VM0042 quantification approaches**: QA1 (Measure & Model) where DNDC/DayCent simulates SOC and GHG fluxes, and QA2 (Measure & Remeasure) where direct measurements drive credit estimates.

Capabilities at a glance

- Map-based project & plot management on Mapbox with multi-layer rendering (polygons, strata, sampling points, baseline control sites, soil texture, climate zones)
- AI-driven sampling plan generation: stratified random sampling, automatic point allocation, MDD/Power Analysis (FAO 2019)
- Repeat sampling cycle automation with two trigger types: **1-year cap from previous cycle** and **end-of-growth-cycle** (pre-planting window)
- Work order generation (PDF + email + MCP token) for external sampling contractors
- Model run console for DNDC and DayCent with live logs, paired baseline+project simulation, and Monte Carlo uncertainty
- Excel ingestion pipeline for the existing CropNut SOC datasheet schema
- Forecast-vs-actual dashboard, VM0042 compliance scoring, alert engine
- Role-based access, MCP token issuance for external contractors, full audit log

Lean philosophy. This document defines *what* needs to be built and *why*, with mockups and field lists. It deliberately stops short of API endpoints, sequence diagrams, and database DDL — those will be authored by the lead developer once the technical stack and CRM are finalized. This keeps initial scoping cost low.

Three-tier delivery roadmap (top-level)

Tier	Scope & rationale
PRIORITY 1 — Immediate (MVP)	The minimum viable layer to start running real cycles end-to-end: Login, Project Map, Plot Details, Sampling Plan Generator (manual + AI proposal), Work Order, MCP Sampler view, Excel ingestion, basic Admin/Roles. Goal: fielded with one farm (Kisima) and one contractor (CropNut) within first release.
PRIORITY 2 — Post-MVP	The intelligence and automation layer: full AI Agent (chat + autonomous proposal), DNDC/DayCent Model Run Console, Monte Carlo uncertainty, Dashboard with forecast vs actuals, advanced alerting, automated cycle triggers.
PRIORITY 3 — Later	Optimization & advanced features: DayCent recalibration loop, IME workflow, automated MVR draft generation, multi-tenant support, advanced reporting, CRM integration, predictive ML on sampling cost/duration.

3. Personas & roles

Super Admin — Nitzan

Single user · full system access · accountable for org-level decisions

- Creates new projects, configures default templates, manages user access & roles
- Approves baseline scenarios, reviews and signs off on MVRs (with IME)
- Has override authority on AI Agent decisions and can replay any audit log entry
- Manages MCP tokens, model licenses (DNDC/DayCent), AWS storage and integration keys

AI-MRV Agent — service account

Always-on automated user · runs as a sandboxed service identity · connects via API + MCP

- **Proposes** sampling plans (point allocation, MDD calculation, stratification logic) for MRV Manager approval
- **Executes** approved tasks: running DNDC/DayCent simulations, generating work orders, ingesting lab Excels, recalibrating models
- **Chats** with users in natural language ("how many samples does Stratum C need next cycle?", "rerun DNDC with the latest parameter set")
- **Flags** compliance gaps, statistical anomalies (high CV, out-of-window sampling), missing data
- Per-action policy: AUTO / CONFIRM / OFF — see Admin screen, default is CONFIRM for any task that creates or modifies an artifact

Operator 1 — MRV Manager

Internal Carbonature staff · assigned to one or more projects

- Reviews and approves AI Agent proposals (sampling plans, work orders, model runs)
- Issues work orders to external samplers, monitors campaign progress in real time
- Ingests and validates lab Excels returning from CropNut and other partners
- Initiates DNDC/DayCent runs (true-up, verification), reviews uncertainty deductions
- Cannot manage users or change global system configuration

Operator 2 — Soil Sampler

External contractor (e.g., CropNut Kenya) · accesses platform via MCP-issued token

- Receives work orders via email + one-tap MCP activation token
- Connects field tooling (GPS, barcode scanner, camera) to the Sampler View through an MCP server
- For each sampling point: capture GPS, scan barcode of sample bag, photograph bag, add notes
- Cannot edit the sampling plan, cannot view other projects, cannot see lab results
- Token-scoped access: limited to the specific work order and its 37 sampling points

4. High-level architecture

Stack decisions (locked)

Layer	Choice	Notes
Cloud	AWS (eu-west-1)	S3 for assets/Excels, RDS PostgreSQL+PostGIS for spatial data, Lambda for triggers, ECS/Fargate for model runners
Web client	React + TypeScript	Tailwind or styled-components; component library at developer's discretion
Map engine	Mapbox GL JS	Existing Carbonature license. Custom polygon styling matches the green palette of these mockups.
API layer	REST + GraphQL hybrid (developer's call)	Authentication via SSO (Google Workspace) + MCP tokens for external contractors
Spatial DB	PostgreSQL + PostGIS	Polygons, points, strata, samples — all geo-indexed

Object storage	S3 (private buckets)	Excels, lab PDFs, photos, model run outputs, MVRs
AI Agent runtime	Anthropic Claude (recommended) or equivalent LLM with tool use	MCP server hosts both the field-sampler tools and the model-runner tools so the agent can call them uniformly
Model wrappers	Containerized DNDC and DayCent CLI	Carbonature buys licenses; module orchestrates runs via container-based wrappers; logs streamed back to UI
CRM	To be selected	Recommendations in §10

Logical components

- **Web frontend** — 9 main screens (this document), Mapbox-driven, real-time updates via WebSocket for active model runs and sampling progress
- **API service** — auth, projects, plots, strata, sampling points, work orders, lab data ingestion, audit log
- **Spatial DB** — PostGIS, indexed on polygon and point geometries
- **AI Agent service** — Claude-based agent with tool use; connects to Model Wrappers and Sampler MCP via Anthropic's MCP standard
- **MCP Server (Sampler tools)** — exposes `capture_gps`, `scan_barcode`, `capture_photo`, `submit_point` tools to the agent and to the sampler's web-based MCP client
- **MCP Server (Model tools)** — exposes `run_dndc`, `run_daycent`, `recalibrate`, `monte_carlo` tools
- **Excel ingestion pipeline** — IMAP-watched mailbox + S3 trigger + parser that maps the CropNut SOC datasheet to the canonical SOC schema (see §11)
- **Model runners** — containerized DNDC and DayCent CLI wrappers, fed inputs from PostGIS & lab data, outputs streamed to S3 + DB
- **External sources** — Mapbox tiles, SoilGrids API, FAO Soils Portal, PRISM weather, optional CRM

Why MCP? The Model Context Protocol gives one consistent contract for the AI Agent to talk to (a) field tools used by the human sampler and (b) carbon-model wrappers run on the server. The same protocol surface lets the developer add new tools (e.g., "fetch SoilGrids texture") without changing the agent prompt or core API. It also gives external contractors a token-scoped, audit-friendly surface for in-field capture.

5. Topic priority backlog (3 tiers)

This is the canonical scoping table. Each capability the developer will build is mapped to one of three priority tiers, per Carbonature's request to deliver an MVP fast and iterate.

#	Capability	Priority	Rationale & dependency
1	SSO login + workspace overview (1 project)	P1 — Immediate	Without auth nothing else can ship. Single-tenant for v1.
2	Project Map (Mapbox) — polygons, points, strata, layers	P1 — Immediate	Core to all visualization. Layers can be added incrementally.
3	Plot Details panel (history, metadata, sampling lifecycle)	P1 — Immediate	Drives the day-to-day MRV Manager workflow.
4	Sampling Plan Generator — manual (rule-based) version	P1 — Immediate	Stratified-random allocation + MDD calculator. AI proposal layer comes in P2.
5	Work Order generation (PDF + email + MCP token)	P1 — Immediate	Lets us start real campaigns with CropNut.
6	MCP Sampler view (web, GPS+barcode+photo capture)	P1 — Immediate	Replaces paper/Excel-only field workflow. Builds on MCP standard.
7	Excel ingestion pipeline (CropNut schema)	P1 — Immediate	The current lab→DB flow. Mapped 1:1 to the existing template.
8	Admin / Users & Roles (basic — 4 roles, manual invite)	P1 — Immediate	Required for security. AI Agent capability scoping in P2.
9	Audit log (basic — who-did-what-when)	P1 — Immediate	Compliance baseline. Verra audits require traceability.
10	AI Agent — proposal mode (sampling plans, alerts)	P2 — Post-MVP	Once the manual flow is stable, the agent layers on top of it.
11	AI Agent — chat interface (natural language Q&A)	P2 — Post-MVP	"Ask AI Agent" floating button. Read-only at first, then write actions.
12	AI Agent — execution mode (runs DNDC/DayCent on instruction)	P2 — Post-MVP	Per-action AUTO/CONFIRM/OFF policy.

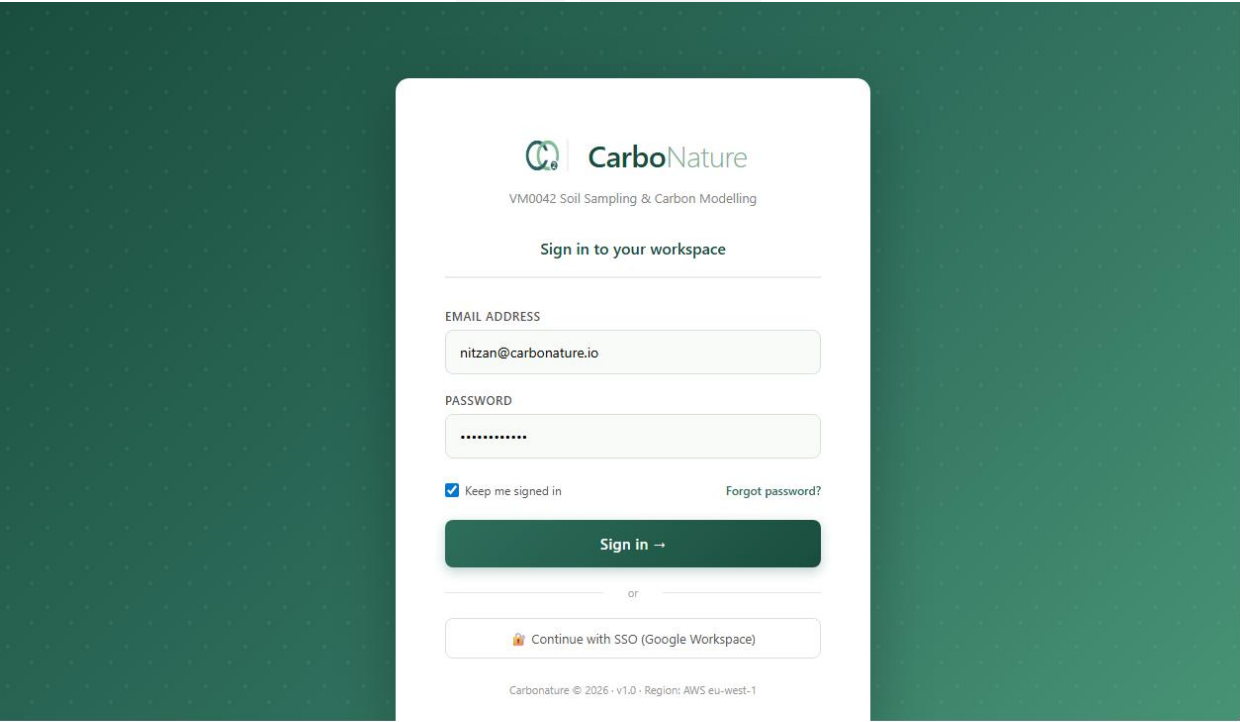
13	Model Run Console — DNDC v9.x	P2 — Post-MVP	Containerized runner + live logs + result preview. Monte Carlo in P3.
14	Model Run Console — DayCent v6.x	P2 — Post-MVP	Same wrapper pattern as DNDC.
15	Dashboard — forecast vs actual	P2 — Post-MVP	Aggregates all P1 + P2 data into a single view.
16	Cycle automation triggers (1-year cap, end-of-growth-cycle)	P2 — Post-MVP	Cron-driven, surfaces drafts to MRV Manager.
17	VM0042 compliance score (rule engine)	P2 — Post-MVP	8 hard checks (≥ 3 control sites, stratified, ≥ 3 composites, ESM, etc.) + warnings.
18	Stratum heatmap and per-stratum statistics	P2 — Post-MVP	Surfaces high-CV strata (Stratum C example).
19	Monte Carlo uncertainty propagation	P3 — Later	VMD0053-aligned. L=500–1000 default. Heavy compute, deferred.
20	Model recalibration / true-up workflow with IME signoff	P3 — Later	Bayesian posterior, MVR generation, IME review portal.
21	Automated MVR draft generation	P3 — Later	Templates from VM0042 webpage; AI Agent fills 80%, MRV Manager finalizes.
22	CRM integration	P3 — Later	Pending CRM selection (see §10).
23	Multi-tenant / partner workspaces	P3 — Later	If Carbonature licenses the platform to other developers.
24	Predictive ML for sampling cost & campaign duration	P3 — Later	Once enough historical campaigns exist (~10 cycles minimum).
25	Native mobile app (iOS/Android) for field sampler	P3 — Later	Web-based MCP view sufficient for v1. Defer until contractors request offline.

6. Functional modules — by screen

Each subsection below covers one of the nine main screens. Mockups are illustrative — all dimensions, colors and spacing are flexible. **Field lists, validation rules, user actions, and priority are binding.**

6.1 Login & Workspace Overview

SCREEN 1 OF 9 · PRIORITY 1 · ROUTES: /LOGIN, /WORKSPACE



Mockup 1 — Login screen with SSO option

Purpose

Single entry point for all internal users. External samplers do **not** log in here — they use a one-tap MCP token from the work order email.

Fields & actions

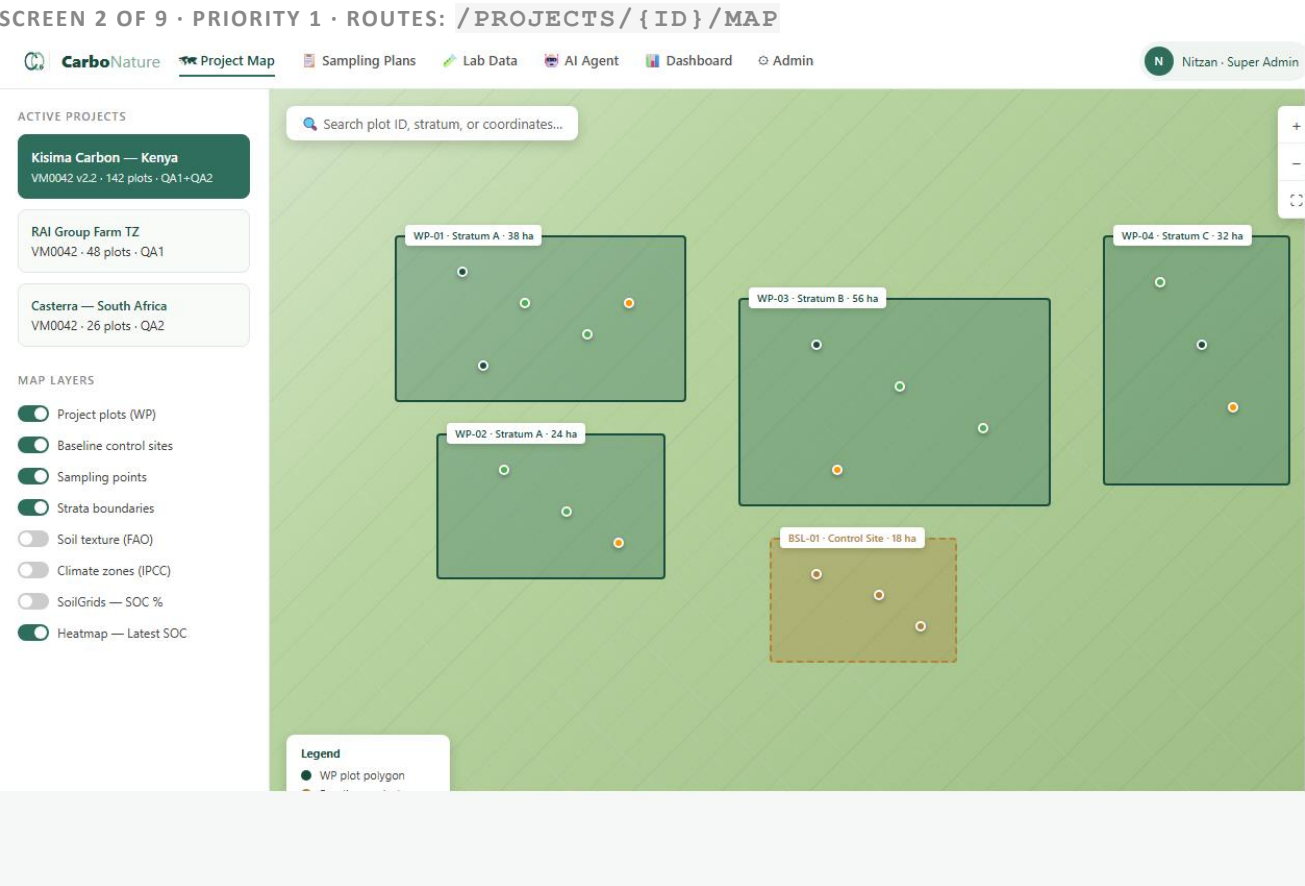
Field	Type	Validation	Notes
Email	email	Required, valid email format, max 100 chars	Domain allow-list configurable in Admin
Password	password	Min 12 chars, complexity per AWS Cognito policy	Hidden when SSO is enabled for the email's domain

Keep me signed in	checkbox	—	Extends refresh token to 30 days
Forgot password	link	—	Email-based reset flow
SSO button	button	—	OAuth2 to Google Workspace; preferred for @carbonature.io

Business rules

- Three failed password attempts → 5-minute lockout. Five → require Super Admin reset.
- SSO is the recommended path for all @carbonature.io users; password is the fallback for guests/auditors.
- Successful login routes to the last-visited workspace, or to Project Map if first login.

6.2 Project Map



Mockup 2 — Project Map with active project, layers panel, and sampling status

Purpose

The primary spatial workspace. Shows all WP polygons, baseline control sites, sampling points (color-coded by status), and toggleable analytical layers. Floating "Ask AI Agent" button is the chat entry point.

Map layers (toggleable)

Layer	Source	Default	Priority
Project plots (WP polygons)	PostGIS	ON	P1
Baseline control sites (BSL)	PostGIS	ON	P1
Sampling points (with status)	PostGIS	ON	P1
Strata boundaries	PostGIS	ON	P1
Soil texture (FAO)	FAO Soils Portal API	OFF	P2
Climate zones (IPCC)	WWF Ecoregions + IPCC	OFF	P2
SoilGrids — SOC %	SoilGrids API	OFF	P2
Heatmap — latest measured SOC	Local data (kriging)	OFF	P3

Sampling-point color codes

- **Dark green** — planned, not yet sampled
- **Light green** — sampled, lab results received
- **Orange** — sampled, lab pending
- **Brown** — baseline control site point

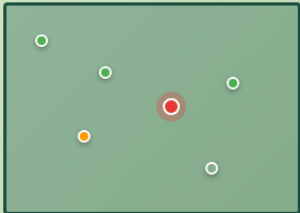
User actions

- Click polygon → opens Plot Details (screen 3)
- Hover sampling point → tooltip with Sample ID, last SOC, status
- Click sampling point → side panel with point history
- Search bar → text search by Plot ID, Stratum, or coordinates (lat/lon)
- "Ask AI Agent" button → opens chat panel anchored to the current map context (P2)

6.3 Plot Details

SCREEN 3 OF 9 · PRIORITY 1 · ROUTES: /PROJECTS/{ID}/PLOTS/{PLOTID}

WP-03 — Stratum B



WP-03 — Sampling Point #14

Plot ID: KIS-WP03-14 · Carbonature ID 3988 · Stratum B · 56 ha

Export CSV

Show on map

Schedule re-sampling

Overview

Sampling history

Lab results

Photos & barcodes

Model runs

COORDINATES

-0.6234, 35.4521

QUANTIFICATION APPROACH

QA1 — Measure & Model

SOIL TEXTURE (FAO)

Clay loam

CLIMATE ZONE (IPCC)

Tropical, dry

SOIL GROUP (WRB)

Vertisol

SLOPE CLASS

Gently sloping (3°)

LINKED BASELINE SITE

BSL-01 (12 km)

LATEST SOC STOCK

42.1 t/ha 2025

SAMPLING STATUS

✓ Sampled · pending lab

SAMPLING LIFECYCLE

✓

Plan generated

2025-01-12

✓

Work order issued

2025-02-03

✓

Field sampled

2025-03-18

⚡

Lab analysis

(in progress)

5

Model true-up

6

Verification

SAMPLING HISTORY (3 CYCLES)

Cycle	Date	Approach	Depth (cm)	SOC (t/ha)	BD (g/cm³)	Lab	Sampler	Sample ID
#1 (Initial)	2020-04-12	QA1	0–15 / 15–30	38.4	1.32 / 1.41	CropNut	J. Mwangi	OFM00000014
#2 (True-up)	2023-04-09	QA1	0–15 / 15–30	40.8	1.31 / 1.39	CropNut	J. Mwangi	OFM00010214
#3 (Verification)	2025-03-18	QA1	0–15 / 15–30	42.1*	1.30 / 1.38	CropNut	P. Kamau	OFM00021614

Mockup 3 — Plot Details with sampling lifecycle timeline and 3-cycle history

Purpose

One pane that gives the MRV Manager and Super Admin everything they need to know about a single plot, sampling point, or stratum. Tabbed: Overview, Sampling history, Lab results, Photos & barcodes, Model runs.

Overview fields (read-only)

Field	Source	Display
Coordinates (lat, lon)	PostGIS centroid (plot) or actual point GPS	6 decimals · WGS84
Quantification approach	Plot config	QA1 or QA2 badge
Soil texture (FAO class)	SoilGrids initial, lab override later	Class name

Climate zone (IPCC)	WWF / IPCC lookup by lat/lon	e.g., "Tropical, dry"
Soil group (WRB)	FAO World Reference Base lookup	e.g., "Vertisol"
Slope class	GIS slope analysis	e.g., "Gently sloping (3°)"
Linked baseline site (QA2 only)	Plot config	BSL-XX + distance in km
Latest SOC stock	Lab + ESM calc	t/ha · year badge
Sampling status	Cycle state machine	"Planned / Sampled / Lab pending / Complete"

Sampling history table

One row per cycle. Columns: *Cycle # · Date · Approach · Depth · SOC (t/ha) · BD (g/cm³) · Lab · Sampler · Sample ID*. Latest cycle marked with asterisk if results are still preliminary.

CarboNature

Kisima Carbon Kenya / New Sampling Plan — Cycle #4

 N Nitzan · Super Admin

Approach	QA1 (DNDC) / QA1 (DayCent) / QA2	per-project	Methodology Section 8
Depth scheme	0–15 / 15–30 cm (default)	0–15 / 15–30	≥30 cm to bedrock; 2 increments
Confidence (α)	0.90 / 0.95	0.90	FAO 2019 eq. 1
Power ($1-\beta$)	0.80 / 0.90	0.80	FAO 2019 eq. 2
MDD target	numeric (t SOC/ha)	0.50	Project-specific
Re-visit existing points	checkbox	ON	ESM continuity
Same-season window	checkbox	ON	VM0042 §8.2.1.1
Composite samples for soil composition (BSL setup)	checkbox	OFF	Initial cycle for BSL definition only

System computes (auto)

- **Stratification check** — every quantification unit divided into homogeneous strata; flags violations
- **n per stratum** — power analysis on historical σ ; minimum 3–5 composites/stratum (VM0042 §8.2.1.2)
- **Point allocation** — random points within each stratum, area-weighted, with optional re-visit of historical georeferenced points
- **Compliance pills** — green: stratified random ✓, ≥3 composites/stratum ✓, ESM 2 increments ✓, BSL within 250 km ✓ (QA2 only); amber: high CV warnings

AI Agent overlay (P2)

Renders a recommendation card with reasoning ("Stratum C is heterogeneous, CV = 38%, agent suggests +1 sample"). Manager can accept, reject with rationale, or open chat to dig deeper. All accept/reject decisions are written to the audit log with the agent's exact reasoning.

Cycle-#1 special behavior

Per Carbonature requirement: **the first cycle includes more sampling points than subsequent cycles**, plus additional soil-composition samples used to define the baseline (BSL) plot. The system flags Cycle #1 with an "Initial — extended sampling" badge and pre-fills "Composite samples for soil composition" = ON.

6.5 Work Order

SCREEN 5 OF 9 · PRIORITY 1 · ROUTES: `/PROJECTS/{ID}/WORK-ORDERS/{WOID}`



CarboNature

Kisima Carbon Kenya / Sampling Plans / Work Order WO-2026-0042



Maya Cohen · MRV Manager

OPEN WORK ORDERS

WO-2026-0042
Kisima — Cycle #4 True-up
37 points

Draft

WO-2026-0038
RAI Group — Initial
22 points

Sent

WO-2026-0029
Casterra — Verification
14 points

In progress

WO-2025-0214
Kisima — Cycle #3
34 points

Completed

WORK ORDER — WO-2026-0042

Soil Sampling Campaign — Kisima Carbon Project

Cycle #4 (True-up) · QA1 / DNDC · 37 sampling points across 4 strata + BSL-01

Download
PDF

Email
contractor

Send to
sampler

ASSIGNMENT DETAILS

CONTRACTOR
CropNut Kenya Ltd. — Patrick Kamau

SAMPLING WINDOW
2026-04-15 — 2026-05-12

LAB DESTINATION
CropNut Lab Nairobi (ISO/IEC 17025)

PROJECT LEAD
Maya Cohen — MRV Manager

APPROACH
QA1 — Measure & Model (DNDC)

DEPTH SCHEME
0–15 cm and 15–30 cm (ESM)

SAMPLING POINTS (SHOWING 6 OF 37)

#	SAMPLE ID	STRATUM	WP/BSL	LAT / LON	DEPTH	COMPOSITE	RE-VISIT
1	OFM00021601	A	WP-01	-0.6201, 35.4498	0–15 / 15–30	5 cores	✓ Cycle #3 point
2	OFM00021602	A	WP-01	-0.6212, 35.4509	0–15 / 15–30	5 cores	✓ Cycle #3 point
3	OFM00021603	B	WP-02	-0.6234, 35.4521	0–15 / 15–30	5 cores	✓ Cycle #3 point
4	OFM00021604	C	WP-03	-0.6256, 35.4540	0–15 / 15–30	5 cores	+ New point
5	OFM00021605	D	WP-04	-0.6271, 35.4555	0–15 / 15–30	5 cores	✓ Cycle #3 point
6	OFM00021606	BSL	BSL-01	-0.6342, 35.4612	0–15 / 15–30	5 cores	✓ Cycle #3 point

FIELD PROTOCOL CHECKLIST

- ☐ Clear all surface organic material before sampling
- ☐ 5 cores per composite, fully homogenized in field, sieved < 2 mm

Mockup 5 — Work Order: list left, full document right with QR-style MCP token

Purpose

The bridge from a planned cycle to a contractor in the field. Generates a printable PDF, an email, and a one-tap MCP activation token. Contractor accepts via the token; from that point on, every action they perform in the Sampler View is tied to this work order.

Sections in the document

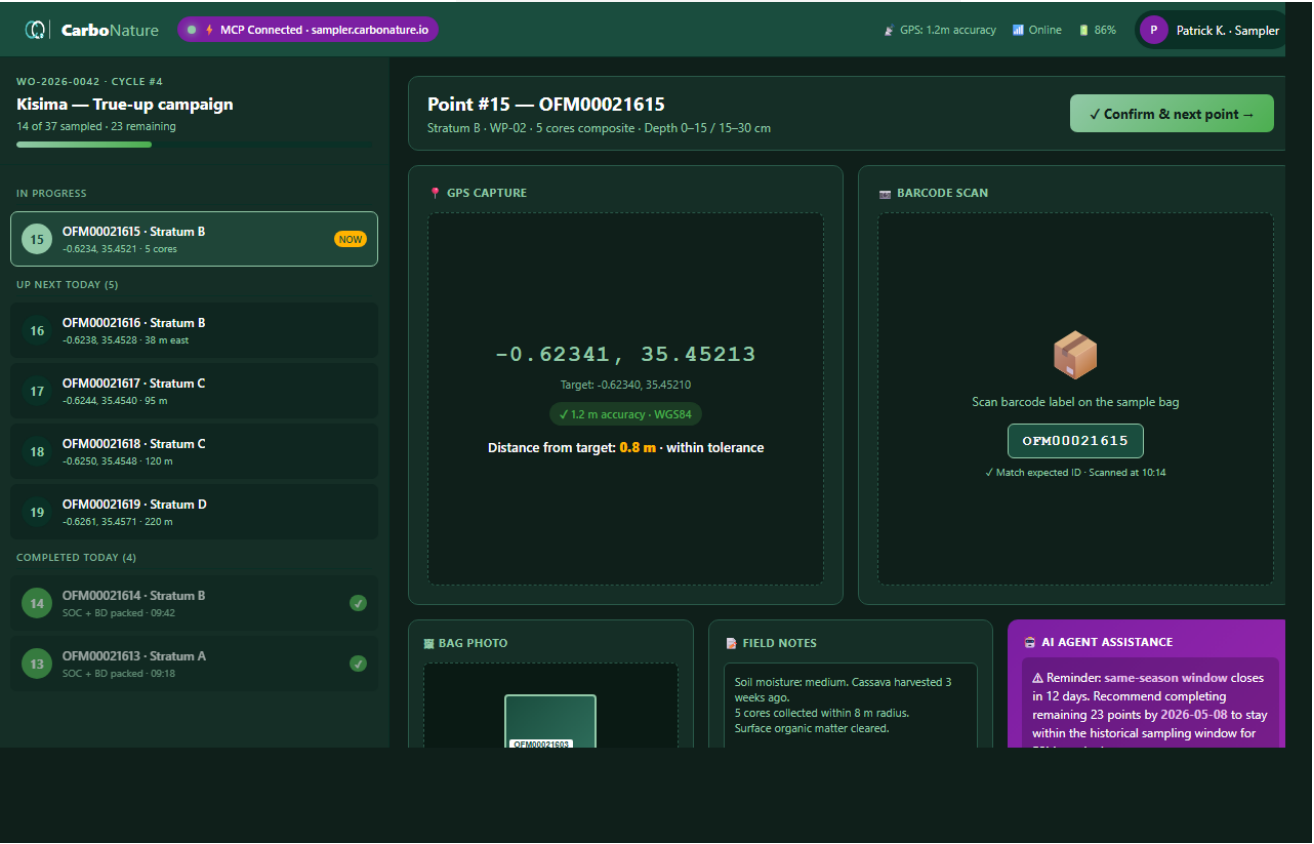
- Header** — WO ID, project name, cycle number, approach
- Assignment** — contractor, sampling window, lab destination, project lead
- Sampling points table** — Sample ID, Stratum, WP/BSL, Lat/Lon, Depth, Composite (n cores), Re-visit flag
- Field protocol checklist** — VM0042 §8.2.1.3 derived (clear surface, sieve <2 mm, photograph, ship within 5 days, etc.)
- MCP connection block** — QR/token + URL
`mcp://sampler.carbonature.io/wo/{woId}` with expiry
- Signatures** — Issued by (MRV Manager), Accepted by (Sampler), Approved by (Super Admin)

States

- Draft → Sent → In progress → Completed → Closed
- Each state transition writes to audit log; "Sent" triggers contractor email + token issuance

6.6 MCP Sampler View

SCREEN 6 OF 9 · PRIORITY 1 · ROUTES: /SAMPLER/WO/{WOID}?TOKEN=...



Mockup 6 — MCP-connected Sampler View: list of points left, capture (GPS + barcode + photo + notes) center, AI agent reminder right

Purpose

The web view that an external sampling contractor uses in the field. **Connected to the AI Agent through MCP:** every capture action (GPS, barcode scan, photo) is exposed as an MCP tool that the agent can also call/inspect. The agent provides context-aware reminders ("same-season window closes in 12 days").

MCP tools exposed by the Sampler MCP server

Tool name	Inputs	Output	Used by
<code>list_points</code>	<code>workOrderId</code>	Array of points + status	Sampler view, Agent
<code>capture_gps</code>	<code>pointId</code> , <code>lat</code> , <code>lon</code> , <code>accuracyMeters</code>	Distance from target, validation result	Sampler view
<code>scan_barcode</code>	<code>pointId</code> , <code>code</code>	Match expected ID? (boolean)	Sampler view

<code>capture_photo</code>	pointId, photoBase64	S3 URL, OCR result of barcode visible	Sampler view
<code>add_field_note</code>	pointId, text	OK	Sampler view
<code>submit_point</code>	pointId	Locked — no further edits	Sampler view
<code>get_progress</code>	workOrderId	Sampled/remaining counts	Agent

Validation per point

- GPS distance from target ≤ 25 m (configurable per project) — soft warning if over
- Barcode must match the expected Sample ID exactly (e.g., `OFM00021615`) — hard error if mismatch
- Photo must be present (no submission without it)
- All four captures (GPS, barcode, photo, notes-or-skip) → "Confirm & next point" button enables

Agent reminders surfaced in this view

- "Same-season window closes in N days"
- "Stratum X has Y points remaining and you're closest to point Z"
- "Lab cutoff in 5 days — please ship by date"

SCREEN 7 OF 9 · PRIORITY 2 · ROUTES: /PROJECTS/{ID}/MODEL-RUNS/{RUNID}

Mockup 7 — Model Run Console: configuration + inputs status left, live terminal log + preliminary KPIs + SOC trajectory right

The control surface for orchestrating DNDC and DayCent runs. Inputs are pre-validated; the run executes in a containerized worker (ECS Fargate); logs stream live; preliminary KPIs and the project-vs-baseline SOC trajectory render as the run progresses. Final results write back to PostGIS and S3.

Input	Source	Loaded check
Initial SOC (t=0)	Lab Excels (CropNut datasheet)	≥ all sample IDs for the cycle present
Bulk density	Lab Excels	matches sample IDs
Soil texture	SoilGrids OR lab override	1 value per cell minimum

Climate (daily)	PRISM API	$\geq \text{run-period years} \times 365 \text{ records per cell}$
ALM activities	DB (entered by MRV Manager)	Schedule of activities for each plot
True-up SOC (most recent cycle)	Lab Excels	Fraction loaded indicator (e.g., 23/37)

Run lifecycle states

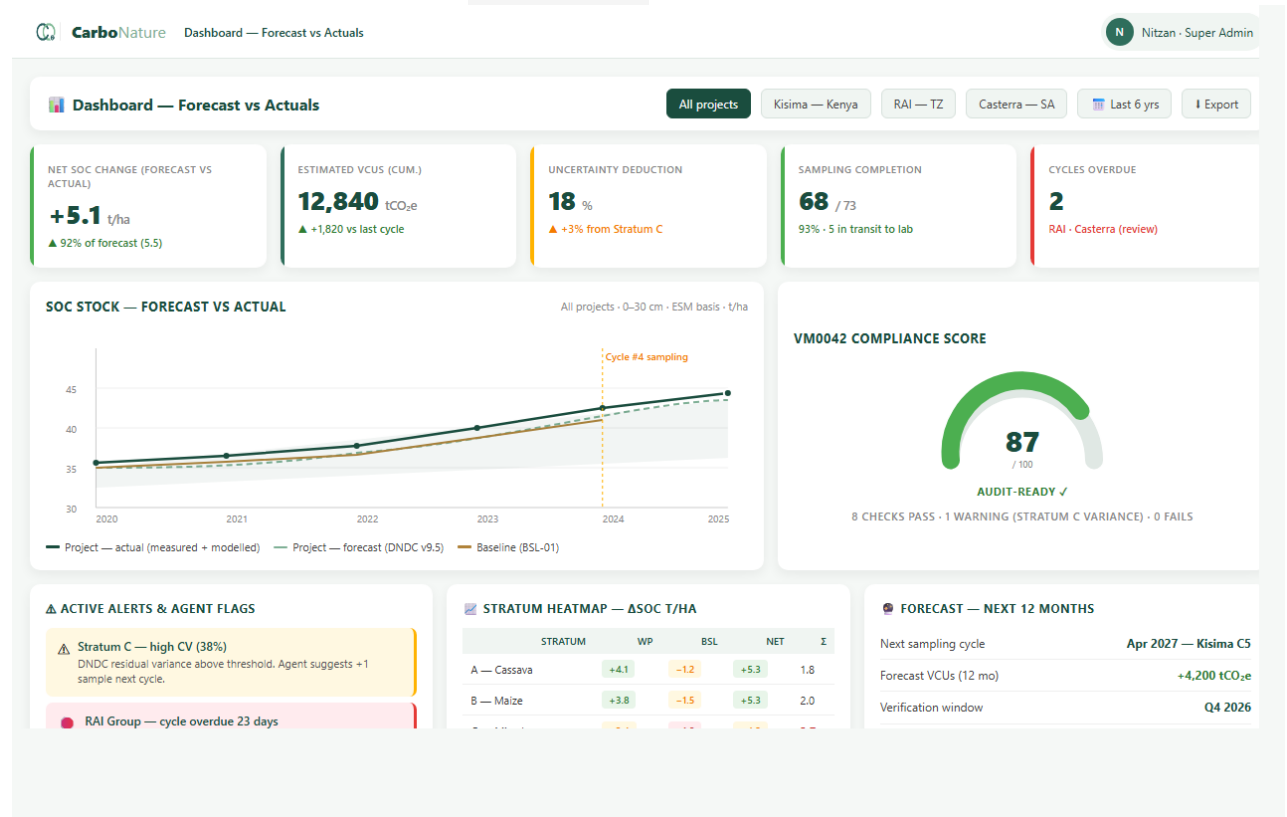
1. **Configuring** — user picks model, scenario, period, scope
2. **Validating** — system checks inputs status; refuses to start if hard inputs missing
3. **Running** — live log streams; baseline scenario, then project scenario, then Monte Carlo (P3) run sequentially
4. **Completed** — KPIs locked, full report written to S3, PDF downloadable, results pushed to Dashboard
5. **Failed** — error captured, retryable; agent assists with diagnosis

KPI preview (live)

- $\Delta\text{SOC}_{\text{wp}}$ (project) — t/ha
- $\Delta\text{SOC}_{\text{bsl}}$ (baseline) — t/ha
- **Net** — t/ha = wp – bsl
- **Uncertainty** — % deduction (P2: simple SE; P3: Monte Carlo)

6.8 Dashboard — Forecast vs Actuals

SCREEN 8 OF 9 · PRIORITY 2 · ROUTES: /DASHBOARD



Mockup 8 — Dashboard: KPIs across top, SOC trajectory + compliance gauge, alerts/heatmap/forecast at bottom

Purpose

The single pane the Super Admin opens daily and the MRV Manager opens before any meeting. Aggregates across all projects (or filters to one). Combines past performance, forecasts, and active risks. **This is the "control tower."**

Top KPIs (5 cards)

1. Net SOC change — actual vs forecast (% of target)
2. Estimated VCUs cumulative (tCO₂e) + delta vs last cycle
3. Uncertainty deduction (%) — current and trend arrow
4. Sampling completion (X/Y points done) — current cycle
5. Cycles overdue (count) — drives the alerts list below

Mid-section

- **SOC stock chart** — three lines: project actual, project forecast (DNDC), baseline. Confidence band on actual. Vertical marker for the most recent sampling cycle.
- **VM0042 compliance gauge** — 0–100 score (rule engine); 8 hard checks + warnings. Color: green ≥80, amber 50–79, red <50.

Bottom row (3 cards)

- **Active alerts & agent flags** — warnings, errors, AI suggestions, each with one-click action
- **Stratum heatmap** — ΔSOC per stratum (WP, BSL, Net, σ); flags strata with high σ

- **12-month forecast** — next sampling cycle, forecast VCUs, verification window, model true-up due, MVR re-submission deadline

6.9 Admin / User Management

SCREEN 9 OF 9 · PRIORITY 1 (BASIC) + 2 (AGENT SCOPING) · ROUTES: /ADMIN/USERS

CarboNature

Admin / Users & Roles

N Nitzan · Super Admin

WORKSPACE

Organization

Users & Roles

SSO & Security

Audit log

CONFIGURATION

Project templates

Lab integrations

MCP servers

AI Agent settings

Model licenses

DATA

Storage (S3)

Import / export

API keys

Users & Roles

5 active users · 2 pending invites · 1 service account (AI Agent)

Export users CSV

Manage MCP tokens

+ Invite user

Users

Roles & permissions

AI Agent identity

External contractors

Search by name, email, role...

All (6)

Super Admin

MRV Manager

Sampler

AI Agent

USER	ROLE	PROJECT ACCESS	STATUS	LAST ACTIVE	AUTH	
NB Nitzan Bauer nitzan@carbonature.io	SUPER ADMIN	All projects	Online	just now	SSO	
AI Carbonature AI Agent agent@carbonature.io · service account	AI AGENT	All projects	Active	continuously	API key + MCP	
MC Maya Cohen maya@carbonature.io	MRV MANAGER	Kisima RAI	Online	2 min ago	SSO	
EB Elad Bouton elad@carbonature.io	MRV MANAGER	Casterra RAI	Offline	2 hours ago	SSO	
PK Patrick Kamau p.kamau@croprnut.co.ke · external	SAMPLER	Kisima	In field	14 min ago	MCP token	
JM John Mwangi j.mwangi@croprnut.co.ke · external	SAMPLER	Kisima RAI	Offline	3 days ago	MCP token	

PERMISSIONS MATRIX

CAPABILITY	SUPER	MRV	AI	SAMPLER
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AI AGENT — CAPABILITY SCOPE

ACTION	AUTO	CONFIRM	OFF
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Mockup 9 — Admin: user list with role badges, permissions matrix, AI Agent capability scoping

Sidebar — admin sections

Section	Function	Priority
Organization	Org-level settings, billing contact, default region	P1
Users & Roles	Invite/revoke users, assign roles & project access	P1
SSO & Security	Configure SSO providers, password policy, MFA	P1
Audit log	Filterable record of all actions; exportable	P1

Project templates	Reusable plan defaults per region/crop type	P2
Lab integrations	Configure Excel mailbox watchers, parser overrides	P1
MCP servers	Manage Sampler MCP and Model MCP endpoints & tokens	P1
AI Agent settings	Per-action AUTO/CONFIRM/OFF policy; LLM provider	P2
Model licenses	DNDC/DayCent license keys, version pinning	P2
Storage (S3)	Bucket assignment, retention policy	P1
Import / export	Bulk import projects/plots, export entire workspace	P2
API keys	Token-based access for integrations	P2

Permissions matrix (P1)

Capability	Super Admin	MRV Manager	AI Agent	Sampler
View map & plots	✓	✓	✓	✓ (own WO only)
Generate sampling plan	✓	✓	Propose	—
Approve work orders	✓	✓	—	—
Capture field data	—	—	—	✓ (own WO)
Run DNDC / DayCent	✓	✓	✓ (with confirm)	—
View lab results	✓	✓	✓	—

Manage users / billing	✓	—	—	—
Export workspace	✓	—	—	—

AI Agent capability scope (P2)

Per-action policy, configurable by Super Admin. Default values shown.

Action	Default	Notes
Propose sampling plan	AUTO	Read-only proposal; manager approves
Send work orders	CONFIRM	Always requires manager click
Run DNDC / DayCent	CONFIRM	Avoids accidental compute spend
Recalibrate model	CONFIRM	Affects all subsequent runs; requires explicit signoff
Issue alerts & flags	AUTO	Read-only; no system changes
Chat with users	AUTO	Read-only by default; write actions still need CONFIRM per type above

7. AI Agent — design & behavior

Three operating modes (all enabled simultaneously)

Mode	What it does	Example
Propose	Generates a draft (sampling plan, alert, flag) for a human to approve	"For Cycle #4, I recommend 37 sampling points across 4 strata, with +1 in Stratum C due to high CV. Approve?"
Execute	Runs an approved task end-to-end (DNDC, DayCent, Excel ingest, work order creation)	"Running DNDC v9.5 with the tropical_drylands_v3 parameter set... complete in 14 min."

Chat	Answers free-form questions and triggers tasks by request	User: "Why is Stratum C uncertain?" → Agent: "CV = 38%, $\sigma = 3.7$. Driver: heterogeneous soil texture. Suggest extra calibration draw on next true-up."
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Implementation guidance

- **LLM** — Claude (Anthropic) recommended given native MCP support and tool-use reliability for agentic workflows; final pick at developer discretion
- **Tool surface** — exposed via MCP servers (Sampler tools + Model tools); the agent's prompt is identical regardless of who initiates the conversation
- **Memory** — short-term per session + long-term per-project knowledge (sampling history, parameter sets, model versions). Persisted in PostgreSQL with embeddings for retrieval.
- **Guardrails**
 - Cannot modify data without explicit MRV Manager approval (CONFIRM gate)
 - Cannot send external emails without approval
 - Every action is logged with full reasoning chain for audit replay
 - Refuses anything outside agronomy / SOC / VM0042 / project-management scope
- **Audit replay** — every agent decision logs: prompt, tool calls, tool results, final action, user response. Super Admin can replay any session.

Sample agent behaviors (priority-tagged)

- **P2** Detect overdue sampling cycles and draft work orders proactively
- **P2** Flag strata with CV above a threshold and suggest sample-count adjustments
- **P2** Auto-validate VM0042 compliance and surface gaps before audit
- **P3** Recommend DayCent recalibration when new parameter sets are published
- **P3** Draft sections of the MVR (model validation report) from run results

8. MCP integration — soil sampler

The Sampler is an external contractor (e.g., CropNut Kenya). They do **not** have a Carbonature SSO account. Instead, when the MRV Manager sends a work order, the system issues a scoped **MCP token**. The contractor opens the URL in any modern browser; the page is the Sampler View (screen 6); under the hood it speaks to the Sampler MCP server.

Token characteristics

- Scoped to a single Work Order (e.g., WO-2026-0042) and its sampling points
- Expires at the end of the sampling window + 7 days (configurable)
- Single-use per session, but the same contractor can request a refresh
- Revocable by Super Admin or MRV Manager from the Admin screen at any time

Why MCP and not just a REST endpoint?

- **Uniform contract:** the AI Agent and the human contractor both use the same tool surface (capture_gps, scan_barcode, etc.). No code drift between human and agent paths.
- **Discoverability:** when Carbonature adds a new tool (say, soil-moisture probe integration), the agent picks it up automatically from the MCP server's tool list.
- **Audit-friendly:** every tool call is logged with the same schema, which simplifies the audit log.
- **Future-proof:** native mobile app (P3) would expose the same MCP tools — zero backend rework.

9. CRM recommendations

Carbonature is in the process of selecting a CRM. Below are the three options most aligned with the platform's needs (project-centric, agronomy-friendly, AWS-friendly) — ordered by recommended fit.

Option	Strengths for Carbonature	Tradeoffs	Pricing band (per user/mo)
HubSpot CRM (recommended)	Generous free tier; strong contact + deal pipelines for project lifecycle (lead → MoU → sampling → verification → VCU sale); excellent native integrations; clean API; works well with our stage-gated sampling model	Less custom-object flexibility than Salesforce; reporting is good but not enterprise-grade	Free → \$90 (Pro)
Salesforce Sales Cloud	Industry standard; deep custom objects for projects/strata; large ecosystem of agronomy add-ons; AWS-friendly via AppExchange	Steep learning curve; expensive; over-engineered for current org size	\$165–330
Pipedrive	Cheapest; very simple deal pipeline; fast onboarding; sufficient for the project-management aspects	Limited custom fields; weaker reporting; would likely outgrow it within 18 months	\$24–99

Recommendation. Start with **HubSpot CRM** on the free or Starter tier (\$20/user/mo). Map the carbon-project lifecycle (Lead → MoU signed → Baseline established → Cycle 1 → Verification → VCU issuance → Sale) as a custom deal pipeline. Integrate with the platform via webhooks: when a Work Order completes, a HubSpot deal stage updates automatically. Re-evaluate at 12 months — if Carbonature crosses ~10 active projects, evaluate Salesforce.

10. VM0042 compliance hooks

The platform should make compliance not just possible but **automatic and visible**. The compliance score (Dashboard, screen 8) is computed from the rules below.

Hard checks (must pass)

1. Stratified random sampling employed for every cycle
2. ≥ 3 composite samples per stratum
3. ESM-based reporting (≥ 2 depth increments)
4. Same-season sampling window respected
5. For QA2: ≥ 3 baseline control sites; each within 250 km; each meeting all 9 similarity criteria from VM0042 Table 7
6. For QA1: model run validated per VMD0053; MVR present and IME-signed
7. Lab is ISO/IEC 17025 or NAPT-participant or GLOSOLAN-registered
8. Dry combustion (Dumas) used unless project-specific deviation is documented

Soft warnings (raise but don't block)

- Per-stratum CV > 30% (suggests +1 sample next cycle)
- Model true-up due within 6 months
- Verification event approaching with >5% sampling shortfall
- Excel parsing produced any null values for required columns

11. Data model — canonical SOC schema (from CropNut Excel)

The existing CropNut SOC datasheet is the contract for lab → DB ingestion. The canonical fields below are derived directly from the supplied `SOC Datasheet table.xlsx`.

Excel column	Canonical field	Type	Notes
no	row_index	int	Row number within the sheet (used for UX, not stored as primary key)
Sample ID	sample_id	string (PK)	Format: <code>0FM00000001</code> (10 digits zero-padded). Carbonature-issued.
Installation	installation_name	string	Project / farm name (e.g., "RAI group farm TZ")
Project	project_id (FK)	string	Carbonature project ID (e.g., "CarboNature I.D 3988")

Date	sampling_date	date	ISO format. Same as field-capture timestamp.
Stratum	stratum_code	string	e.g., "1", "2", "Avtah", "Maoz Haim"
Depth	depth_cm	int (15 or 30)	Lower bound of increment. Two rows per Sample ID (15 + 30) → 0–15 and 15–30 cm.
BSL/PR	scenario	enum: BSL / PR / WP	Identifies baseline vs project-with sample
BD [gr/cm³]	bulk_density	float	Per increment
SOC [t/ha]	soc_t_per_ha	float (computed)	Excel formula: $(\text{TOC}\% / 100) \times \text{BD} \times \text{Depth} \times 1000$. Replicate in service layer; do not depend on Excel formula evaluation.
TOC [%]	toc_pct	float	Total Organic Carbon — primary lab measurement
TIC 900 [%]	tic_900_pct	float	Total Inorganic Carbon at 900°C
TC [%]	tc_pct	float	Total Carbon
N [%]	n_pct	float	Total Nitrogen
C/N ratio	cn_ratio	float	Computed C/N
ROC 600 [%]	roc_600_pct	float	Recalcitrant Organic Carbon
TOC 400 [%]	toc_400_pct	float	TOC at 400°C

Sheet 2 — Sampling dashboard (existing)

The second sheet of the workbook holds a pivot summary: *Sampled points / Total points / Average SOC* by stratum and by BSL/PR. This is replaced by the platform's Dashboard (screen 8) and need not

be ingested separately, but the parser should preserve the source workbook in S3 for audit traceability.

Ingestion flow

1. Lab emails Excel to `labs@carbonature.io`
2. IMAP watcher saves the attachment to `s3://carbonature-labs/incoming/{date}/{filename}.xlsx`
3. S3 trigger fires a Lambda that runs the parser
4. Parser maps to the canonical schema, validates against expected Sample IDs (from open Work Orders), writes to PostgreSQL, and links to plot/stratum
5. On success: status flag flips on the Map (orange → green), Dashboard refreshes, AI Agent is notified
6. On failure: parsing errors written to a quarantine table; MRV Manager sees an alert with row-level diagnostics

12. Non-functional requirements (lean)

Aspect	Target	Notes
Availability	99.5% (business hours)	Platform is workflow-driven, not real-time critical
Map render	<2 sec for 500-point project	Mapbox vector tiles, polygon simplification at low zoom
Excel ingestion	<30 sec end-to-end for <100 rows	S3 trigger + Lambda
DNDC run	≤30 min for 150-plot project · 6 yrs	Containerized in ECS Fargate
Concurrent users	~20 (initial)	Single-tenant; scales horizontally with API replicas
Storage retention	10+ years	VM0042 audit trail; S3 lifecycle to Glacier after 2 yrs
Security	Encryption at rest (AWS KMS) and in transit (TLS 1.3)	SSO + MFA mandatory for internal users

Audit log retention	Project lifetime + 5 years	Append-only PostgreSQL table; mirrored to S3 (Glacier)
PII handling	Sampler email + name only; no farmer-PII stored	Farmer details live in the future CRM, not in the platform
i18n	English-only v1	Hebrew UI deferred to P3; field-tooling supports any locale via MCP tool input

13. Acceptance criteria — MVP (P1)

1. Super Admin can log in via SSO and see all 3 active Carbonature projects on the map.
2. MRV Manager can open a project, click any plot, and see a populated Plot Details panel.
3. MRV Manager can generate a sampling plan for a project (manual rules, no AI required), see the plan on a preview map, and approve it.
4. Approving a plan creates a Work Order with a downloadable PDF, an email to the contractor, and an MCP token URL.
5. An external sampler can open the MCP token URL, see the list of points, and capture GPS + barcode + photo + notes for at least one point.
6. The CropNut Excel file (`SOC Datasheet table.xlsx`) lands in `labs@carbonature.io` and within 1 minute the parsed rows appear under the project's Lab Data tab and on the map (status flips to green).
7. Super Admin can invite a new MRV Manager, scope them to one project, and revoke an MCP token.
8. Audit log contains entries for each of the above actions with timestamp, user, action, and target object.

14. Out of scope (v1)

- Native mobile apps (iOS/Android) — web MCP view is sufficient for v1; defer to P3.
- Lab Results Upload & QA/QC dedicated screen — covered implicitly by the Excel ingestion pipeline (no manual upload UI in v1).
- Compliance Report generator — the Dashboard's compliance score covers the v1 need; full audit-ready PDF deferred to P3 (with MVR generator).
- Multi-tenant / partner workspaces.
- Hebrew UI (system labels). Field input can be in any language.
- Direct Verra Registry submission — manual export to Verra still required in v1.
- Biochar, livestock manure-management, or other VM0042 add-on modules.

15. Appendix A — VM0042 reference checklist for the developer

The platform's compliance score and rule engine should validate against this checklist. Each rule maps to a specific section of VM0042 v2.2.

Rule	VM0042 section	Where enforced in platform
Stratified random sampling	§8.2.1.2	Sampling Plan Generator
≥3–5 composite samples per stratum	§8.2.1.2	Sampling Plan Generator
≥3 baseline control sites (QA2)	§8.2.1.2	Project setup
BSL within 250 km of WP (QA2)	§8.2.1.2 Table 7	Project setup
9 similarity criteria for BSL (QA2)	§8.2.1.2 Table 7	Project setup
≥30 cm depth · ≥2 increments	§8.2.1.3	Sampling Plan Generator
ESM-based SOC reporting	§8.2.1.6	SOC computation service
Sampling within 5-day shipping window	§8.2.1.3	Sampler view + work order
Same-season sampling	§8.2.1.1	Sampling Plan Generator + agent reminders
Dry Combustion (Dumas) preferred	§8.2.1.4	Lab integration config
ISO/IEC 17025 lab	§8.2.1.4	Admin / Lab integrations
NAPT or GLOSOLAN participation	§8.2.1.4	Admin / Lab integrations

5 model selection criteria (QA1)	§4.4 a–e	Model Run Console
5-year true-up cap	§8.2.1, §8.6.1.3	Cycle automation triggers
VMD0053 model validation	VMD0053 §5.2	Model Run Console + MVR (P3)
IME signoff on MVR (QA1)	VMD0053 Appendix 1	P3 — MVR generator

End of specification — Carbonature AI-Driven Soil Sampling & Carbon Modelling Module · v1.0

This document supersedes the initial Hebrew specification. All figures are illustrative.

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